

# Physically-bounded litter-input priors

Literature basis and design choices for the `sigma_input` re-parameterisation (HIKET)

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14 August 2026

## ADDENDUM 2026-08-14 — the ceiling in this note is anchored on the wrong statistic

Everything below documents the original design and remains the record of why `flux_pair` exists and what it fixed. This addendum supersedes only the choice of the ceiling. Decided, **not yet implemented**; full write-up with tables in `manuscript/M&M_parameterization_working_document.tex`, section “Prior specification: the litter-input flux window”.

**The defect.** `sigma_input` is a single global scalar, so `sigma_input × J` is a **national mean** effective flux. The ceiling of 8.7 tC/ha/yr adopted below is Gower et al. (2001)’s Class I evergreen **maximum** (912 gC m<sup>-2</sup> yr<sup>-1</sup>), i.e. the most productive single stand in a *global* boreal compilation. **A mean has been bounded by a maximum**, which is why the constraint has never bound: posterior effective fluxes are 4.99–6.56 tC/ha/yr, all inside. The *direction* of the bound is sound (`litter` NPP is an identity); the statistic and the population are not.

**Unit trap.** Gower et al. (2001) reports in gC m<sup>-2</sup> yr<sup>-1</sup> (Table 4 caption). Zheng et al. (2004) reports in **dry matter**. Proof is internal: Zheng cites Gower’s world-boreal TNPP as 109–1827 (mean 892) where Gower’s own Appendix A gives 218–912 (mean 424) gC — a ratio of exactly 2.00–2.10. Zheng’s headline 563 is therefore **2.81**, not 5.63, tC/ha/yr. Any comparison mixing the two is out by 2×.

**The Finland-relevant numbers** (all understorey-inclusive, carbon units):

| source                                       | quantity                                      | tC/ha/yr         |
|----------------------------------------------|-----------------------------------------------|------------------|
| Tupek, Zenodo<br>10.5281/zenodo.19736499     | tree litter <b>J</b> ( <b>no</b> understorey) | 2.51             |
| Lehtonen & Heikkinen (2015)                  | total litter                                  | 2.70             |
| Zheng et al. (2004) gridded,<br>FI+SE (DM→C) | total NPP                                     | ~2.81            |
| Gower et al. (2001) <b>Nordic</b><br>Class I | total NPP                                     | 3.21 (2.15–4.62) |
| Gower et al. (2001) Class I<br>evergreen     | total NPP                                     | 3.87 (2.14–9.12) |
| <b>ours, calibrated</b>                      | <code>sigma_input × J</code>                  | <b>4.99–6.56</b> |

**Decision: narrow the window to [0.05, 4.62]**, anchored on Gower’s **Nordic** Class I maximum (462 gC m<sup>-2</sup> yr<sup>-1</sup>). This fixes the statistic and the population, keeps the project’s “widest defensible” principle (it assumes Finnish forests are as productive as the most productive stand in the Nordic sample), and depends on none of the arguments that failed scrutiny. It gives `sigma_input` 1.84, which binds in all six models. Zheng’s 2.81 was rejected as a ceiling because it lies **below the prior centre** (`sigma_input` = 1.30 flux 3.26), so the centre would not be representable, and because at only 4% above LUKE’s own total litter it amounts to asserting the inventory as exact.

**Narrowing tightens the prior as well as the wall.** Because the transform is a scaled logit onto the window, at fixed `sigma_ppm` = 0.50 the  $\pm 1$  ppm range moves from `sigma_input` [0.93, 1.72] to [1.09, 1.47]. Decide explicitly whether that is wanted, or whether `sigma_ppm` should rise to compensate.

**Three claims that did NOT survive scrutiny — do not revive them.** (1) “our flux is physically impossible” — it exceeds the Nordic *mean* but not the Nordic *range*; (2) “LUKE’s litter product is biased high” — its 2.70 is 84% of Gower’s Nordic mean, but that range (2.15–4.62) is too wide to support the inference; (3) “Finnish forests are 2× Gower’s productivity” — this compared Korhonen’s *current annual increment* with Gower’s *mean annual increment* (biomass ÷ stand age, stands averaging 99 years); the factor is largely definitional, and Gower’s stands are selected for having complete NPP budgets, not for representing Finnish managed forest.

**What J actually contains** (Zenodo deposit + the YaYasso implementation of the LUKE chain): it is a **flux** (biomass × turnover, Liski et al. 2006 rates); it **includes harvest residues and natural mortality** — both previously assumed missing — and **excludes understorey**, which is what the 1.30 centre stands in for. Legend: `nwl` = foliage + fine roots, `fwl` = branches + coarse roots + stem/bark, `cwl` = stumps. **Open question for B. Tupek:** which fine-root biomass model and turnover rate were used? YaYasso variants span foliage × 0.18–2.5 (biomass) and × 0.5–0.85 (turnover) — a ~20-fold range in the dominant `nwl` component, and exactly the term Lehtonen & Heikkinen exclude (“The uncertainty in these leaf mass-to-fine root ratios was not included in our analyses”), which makes their  $\pm 10\%$  a floor. One variant (`fineroot.total.tsum`) is annotated as *including understorey*, which would make the 1.30 centre a partial double-count.

**Sequencing.** Wait for Roihu jobs 654390–654392 (Yasso07/15/20, corrected target), analyse, then decide. Do not bundle with the `sigma_input` prior tightening.

Appendix diagnostic figure: `manuscript/figures/SX_input_vs_literature.png` ( **not for production** — several comparisons on it are deliberately not like-for-like).

# 1 Summary

`sigma_input` multiplies the litter forcing every model integrates. Under the current homogenised prior it is a weak nuisance parameter, and the likelihood drags two models (TP2, TP3) to effective litter fluxes of 34–50 tC/ha/yr — above the net primary production of *any* forest biome on Earth. This note records the literature basis for replacing that weak prior with a **physical bound on the effective litter flux**, homogeneous across all six models, and the design choices taken. It feeds the six `Prior_specs/*_priors.R` and the manuscript methods.

**Decision:** bound the effective flux to the *widest defensible boreal* window,  $\text{sigma\_input} \times J \in [0.5, 8.7]$  tC/ha/yr, anchored on the Gower et al. (2001) boreal total-NPP range.

## 2 The problem

The effective input flux is  $\text{sigma\_input} \times J$ , where  $J$  is the tree-litter object shared by all six models (`Data/LitterData/tree_litter_per_site_year_awen*.csv`, Tupek et al.), median  $\approx 2.8$  tC/ha/yr across plots (`nwl + fwl + cwl`: non-woody + fine-woody + coarse-woody litter, in AWEN pools).

The current prior treats `sigma_input` as a weak Tier-3 nuisance (log-transform, centre 1, log-SD  $0.5 \Rightarrow 95\%$  CI  $\approx [0.37, 2.66]$ ). But litter input has a physical referent the weak prior discards, so the posterior blows through it (Table 1; `doublechecks/effective_litter_flux_vs_physical.R`).

Table 2: Posterior-median effective litter flux by model.

| Model      | <code>sigma_input</code> (median) | effective flux (tC/ha/yr) | plausible?                                       |
|------------|-----------------------------------|---------------------------|--------------------------------------------------|
| SP1        | 0.32                              | 0.8                       | yes                                              |
| Yasso07    | 1.4                               | 3.5                       | yes                                              |
| Yasso15    | 1.9                               | 4.8                       | yes                                              |
| Yasso20    | 3.0                               | 7.5                       | yes (very productive edge)                       |
| <b>TP2</b> | 13.6                              | <b>34</b>                 | <b>no</b> — 4× the most productive boreal forest |
| <b>TP3</b> | ~20                               | <b>50</b>                 | <b>no</b> — 6×                                   |

TP2/TP3 buy their (best-in-study)  $R^2$  by manufacturing carbon — the same root cause as the TP3 interannual oscillation (a cascade-starved fast pool force-fed to reach the observed stock). `sigma_input`’s distance from  $\sim 1$  is really a **structural-adequacy gauge**, interpretable only once the prior encodes the physical ceiling.

## 3 What `sigma_input` legitimately represents

$J$  is **tree litter** covering the full living-tree biomass-component turnover: foliage/needles + **fine roots** + branches + **coarse roots** + stem bark + stumps. So **belowground (root) litter**

**IS included** (confirmed by B. Tupek and the dataset size-class table: **nwl**=foliage+fine roots, **fwl**=branches+coarse roots+stem bark, **cwl**=stumps) — *J* is *not* aboveground-only. Physical terms *not* in *J*, which **sigma\_input** should absorb, in order of magnitude:

1. **Understory / ground-vegetation (incl. moss) litter** — the dominant missing term. In Finnish boreal forest \$15–35% of total litter production (larger at northern, less productive sites): e.g. in a boreal *Betula pendula* stand total soil-C input (tree + understory) was 2.83 tC/ha/yr with understory **35%**  $\approx$  **0.99 tC/ha/yr**. So **sigma\_input**  $\approx$  1.2–1.5 is *physically justified by understory alone*.
2. **Mycorrhizal mycelial turnover / root exudates** — also absent; large but very uncertain, a further upward push.
3. **Aboveground tree mortality (deadwood / CWD from whole-tree death)** — genuinely omitted (the dataset is turnover-based, not demographic), but a *small* soil-input term in managed Finnish forest: harvest removes most stems, natural mortality is suppressed, and deadwood enters measured soil C only slowly and partly. Minor next to understory.
4. **Tree-litter-model bias** — Tupek model error in the tree flux itself.

Neither mechanism can plausibly inflate soil input several-fold: a boreal forest’s *total* detritus input to soil is capped by its net primary production.

## 4 Literature envelope (boreal, widest defensible)

The quantity to bound is **total forest detritus input to soil** (foliage + woody + fine roots + understory), i.e. the effective flux — **not** aboveground litterfall, which most papers report and which is only  $\sim$ half of the total.

**Anchor** — **Gower et al. (2001)**. Boreal total (above- + belowground) NPP ranges **52–868 gC m<sup>-2</sup> yr<sup>-1</sup>**, mean **424  $\rightarrow$  0.52–8.68 tC/ha/yr, mean 4.24**. At steady state a forest’s total litter/detritus input to soil  $\approx$  its NPP (all production eventually becomes detritus). NPP is in fact a **generous upper bound** on soil input, because in managed Finnish forests harvested stemwood leaves as timber rather than litter — exactly what “as wide as defensible” calls for. Corroborated across the boreal/temperate/tropical global database of Luyssaert et al. (2007).

**Cross-checks** — all comfortably inside [0.52, 8.68]:

- Finnish *Pinus sylvestris* measured tree litter: aboveground 1.31–2.36 + belowground 0.70–0.91 = **2.0–3.3 tC/ha/yr** (belowground 23–37% of total) — matches our *J* median 2.8.
- + understory/moss \$0.5–1.0  $\rightarrow$  typical boreal *total* input  $\approx$  **2.5–4.3 tC/ha/yr**.
- Productive Norway spruce aboveground litterfall up to \$4 tC/ha/yr (central Europe \$8Mgha<sup>-1</sup> biomass; Berg et al. synthesis).
- **Global context:** even the most productive *tropical* forests reach total NPP  $\approx$  12–15 tC/ha/yr. TP2/TP3’s 34–50 tC/ha/yr exceed the NPP of **any** forest biome on Earth by 2–4 $\times$  — categorically impossible, let alone for boreal.

## 5 Chosen bounds

Physical envelope on the effective total-input flux, from Gower et al. (2001) boreal total NPP, applied **homogeneously to all six models and to both ends of the pre-run** — the contemporary

flux *and* the 1917 historical flux:

$$F_{\text{now}}, F_{1917} \in [0.05, 8.7] \text{ tC/ha/yr}$$

where  $F_{\text{now}} = \text{sigma\_input} \times J$  and  $F_{1917} = \text{sigma\_init} \times \text{sigma\_input} \times J$ .

**Ceiling (8.7)** — the physically binding limit, anchored on the Gower boreal total-NPP maximum (widest defensible;  $\text{NPP} \geq \text{actual soil input}$ ). At the plot-median  $J \approx 2.8$  this is a multiplier ceiling  $\text{sigma\_input} \lesssim 3.1$ .

**Floor (0.05)** — deliberately relaxed *below* the Gower minimum (0.52) to a small, strictly-positive guardrail. It is **non-binding** for the global calibration (all effective fluxes land at 0.8–7.5, far above it), so it changes nothing now; it is set low **on purpose**, to pre-position for a future local/stratified **sigma** where disturbed strata (e.g. clear-cuts) genuinely want low flux (see the forward-note). It is kept strictly positive, **not zero**, so the model stays non-degenerate: zero flux  $\Rightarrow$  zero predicted SOC  $\Rightarrow$  a  $-\infty$  likelihood.

**sigma\_init is not bounded on its own.** It has no independent physical referent — it is a *ratio* of two fluxes (how much more or less productive 1917 was than today). Instead we bound the **1917 flux**  $F_{1917}$  to the *same* envelope: a boreal forest in 1917 was subject to the same NPP physics as one today, so Gower is a statement about boreal productivity, not about a particular decade. We then let  $\text{sigma\_init} = F_{1917}/F_{\text{now}}$  fall out. This is a **joint** constraint on the two multipliers, imposed cleanly by re-parameterising in flux space (Appendix). It replaces — and corrects — an earlier idea of an independent guardrail [0.1, 1.5] on **sigma\_init**: that did *not* actually keep the pre-run flux inside the envelope, because  $\text{sigma\_init} \times \text{sigma\_input}$  can reach  $1.5 \times 3.1 \approx 4.6$ , i.e.  $F_{1917}$  up to  $\sim 13$  tC/ha/yr, above the ceiling.

**Ensemble scope — why clear-cuts do not break this.** **sigma\_init** and **sigma\_input** are **single scalar multipliers shared across all plots**, so the envelope constrains the *network-average, typical-forest* flux — exactly where the NPP range is the right statement. Local, transient disturbances (clear-cut, fire, harvest) push *individual* plots below the floor, but they live in the **per-plot**  $J$ , not in the scalar multipliers, so they are represented where they belong and never require the average multiplier to leave the envelope. The envelope is deliberately an *ensemble* prior on average boreal productivity, not a per-plot bound; clear-cutting is precisely why it must stay ensemble-level.

## 5.1 Consequences per model

- **SP1, Yasso07, Yasso15** (0.8 / 3.5 / 4.8): well inside — untouched.
- **Yasso20** (7.5): **within** the widest boreal envelope ( $7.5 < 8.7$ ). At the *widest defensible* bound Yasso20 is a very-productive-but-possible boreal forest, so it is deliberately **not** bound. That Yasso20 sits at the physical edge is itself a reportable result.
- **TP2, TP3** (34 / 50): **bound** at the ceiling. This removes their escape hatch  $\rightarrow$  forces the misfit onto the (now prior-pinned) rates or into a visible  $R^2$  hit = exposes structural inadequacy instead of hiding it.

## 6 Implementation: how to enforce the bound in the sampler

Two questions are easy to conflate but are separate:

1. **What is the bound?** — the physical window  $[0.5, 8.7]$  tC/ha/yr on the effective flux. This is settled (previous sections).
2. **How do we make the sampler respect it?** — the *enforcement mechanism*. The natural idea is a hard boundary in the prior; below we explain why that struggles here, and what we do instead (a re-parameterisation).

## 6.1 A scale factor is unavoidable (and it is not “data” the sampler touches)

The bound is a statement in **physical flux units** (tC/ha/yr). But the parameter the sampler actually moves, `sigma_input`, is a **dimensionless multiplier** — the model applies `sigma_input`  $\times J$  inside each wrapper, and  $J$  (the per-plot litter magnitude) is what carries the units. To state a flux bound as a bound on the multiplier we must divide by the litter scale:

$$\text{sigma\_input} \in \left[ \frac{0.05}{\bar{J}}, \frac{8.7}{\bar{J}} \right] \approx [0.018, 3.1] \quad (\bar{J} \approx 2.8).$$

$\bar{J}$  is a **unit-conversion factor** (an “exchange rate” between flux and multiplier), not a live reading of the calibration data: it is one number, the cross-plot mean litter, known up front and frozen for the whole run. It can be typed in by hand ( $[0.018, 3.1]$ ) or — cleaner — computed once at setup from the litter the engine already loads, so the prior file can state the literature-native physical window  $[0.5, 8.7]$  and the engine does the division. Either way the data flow at run time is unchanged: litter still enters only in the wrapper, exactly as now. This factor is a **units bridge** and nothing more; it is needed however we choose to enforce the bound. What follows is about the enforcement itself.

## 6.2 The natural idea: put a wall in the prior — and why we don’t

The obvious way to add a constraint — and the way a standard Bayesian calibration does it — is to write it straight into the **prior** as a hard boundary: give `sigma_input` zero prior probability anywhere outside the window (a *truncated* prior, or a uniform box with a floor and a ceiling). Everything inside is allowed; everything outside is forbidden.

There is nothing wrong with this *as a description of what we want*. The posterior it defines is exactly the right one. The difficulty is not the destination but the **journey the sampler has to take to it** — and it shows up precisely for the two models the bound exists to catch.

Recall *why* we are doing this: the likelihood is dragging TP2 and TP3 toward absurdly high litter. So their `sigma_input` does not want to rest in the middle of the window — it wants to be pinned **hard against the ceiling**. Now picture the sampler there. At each step it proposes a nearby value and either keeps it or rejects it. Sitting right at the wall, a large fraction of the proposed steps land *outside* the window, where the prior is zero, so they are simply thrown away. The chain keeps getting shoved into the wall and bounced back, taking tiny effective steps, revisiting nearly the same value over and over. Two costs follow. First, the chain explores very slowly — many iterations, few genuinely independent samples (poor mixing). Second, the cloud of samples that piles up right at the boundary is *partly an artefact of the sampler being stuck*: you can no longer cleanly tell “the data want it at the ceiling” from “the sampler could not get away from the ceiling.” And all of this is worst for exactly the parameters we introduced the bound to diagnose. A wall is hardest to sample against precisely where the posterior leans on it.

### 6.3 What we do instead: move the wall out of reach

The re-parameterisation (Option A) keeps the **same target** — the same allowed window, the same posterior — but reaches it without ever letting the sampler touch a wall. The trick is to let the sampler walk on a **re-drawn ruler**.

Instead of sampling `sigma_input` directly (where 0.5 and 8.7 are real places it can walk off the end of), we let it walk on a stretched, endless ruler and simply *read off* the physical value from where it stands. The ruler is marked so that physical values crowd together as you approach either limit: 0.5 is printed infinitely far to the left, 8.7 infinitely far to the right, and the middle of the window sits in the middle of the ruler. The sampler takes ordinary, unobstructed steps along this ruler — there is no end to fall off, so **every proposal it makes is automatically a legal value inside the window**. As it walks far to the right, the physical value it reads glides *toward* 8.7 and slows down, approaching but never reaching it — an asymptote, not a collision.

For the two hard cases this is exactly the behaviour we want. “TP2 wants the maximum litter” stops being “keep slamming into a wall at 8.7” and becomes “keep strolling rightward along an endless ruler” — a smooth, well-mixing walk. The bound is still absolutely respected (the ruler *cannot* read a value outside the window), but it is respected by construction rather than by rejection. The wall has been moved to infinity, so the sampler never spends effort fighting it.

The re-drawn ruler is a smooth **S-shaped map** from the endless coordinate onto the window — the same logistic curve behind an ordinary logit prior, just stretched to span the physical window. We never need its algebra to use it (it is written out in the Appendix); what matters is its shape: steep in the middle, flattening toward each end so the ends lie infinitely far out.

### 6.4 The same ruler, used twice (the two auxiliary parameters)

We apply this trick not once but twice, because there are **two fluxes** that must stay physical: today’s litter input and the 1917 input the pre-run starts from. Each gets its own endless ruler onto the *same* physical window — one reads off the contemporary flux, the other the historical flux. `sigma_input` is recovered from the first; `sigma_init` is simply the **ratio** of the two (how much more or less productive 1917 was than today). So we never bound `sigma_init` directly — we bound the two fluxes it is built from, and it inherits a physical range automatically. Both walks are wall-free in exactly the same way, and because they are two separate rulers the density correction below is just applied to each.

### 6.5 The one catch: an unevenly stretched ruler needs a density correction

This is where the Jacobian comes in, and it is the *only* extra thing Option A asks of us.

Because the new ruler is stretched — packed tightly near the two ends, loose in the middle — an evenly-spread walk along it is **not** evenly spread in physical units. If we did nothing, we would silently be imposing a lopsided prior on `sigma_input` that we never chose: more weight where the ruler is loose, less where it is packed, purely as a side effect of the stretching. That is not a bound problem, it is an accidental prior we did not ask for.

The fix is a single correction term. A probability density is “probability *per unit length*” — so the instant you change what “a unit of length” means along the ruler, you must re-scale the density to match, by multiplying by how much the ruler is locally stretched at each point. **That local stretch factor is the Jacobian**. Including it (in practice, adding its logarithm to the log-prior)

guarantees that the prior we actually get in physical units is the prior we *intended* — a genuinely flat prior across the window, or a mild pull toward 1, whatever we wrote — rather than a by-product of the coordinate change. It is one term, computed automatically alongside the map; no tuning, no interpretive cost.

If it helps: the map re-labels the axis, and the Jacobian is the bookkeeping that keeps the areas under the curve honest after the re-labelling.

## 6.6 In short

Same constraint and same posterior as the naive truncated prior — but reached by walking a wall-free, stretched coordinate instead of rejecting proposals at a wall. We pay exactly one price, the Jacobian correction, and in return the two binding models (TP2/TP3) mix cleanly against a bound they would otherwise sit stuck against.

## 6.7 Remaining caveats

- **Scalar vs per-plot.** `sigma_input` is a single number shared across plots, and the window is placed using the mean litter  $\bar{J}$ , so a handful of high- $J$  plots could nudge just past 8.7. This is second-order relative to killing the 13–20 $\times$  inflation — a gross, everywhere-violated excursion, not a few-plots-over-8.7 effect; noted in methods.
- **sigma\_init.** Not bounded directly: the same wall-free ruler is applied to the 1917 flux, and `sigma_init` is recovered as the ratio of the two fluxes (§“Chosen bounds”, Appendix). Its posteriors already sit interior (0.19–0.72), so the envelope is non-binding for it — construction stays uniform, behaviour unchanged.
- **Floor relaxed to 0.05, not 0.** Non-binding now; kept strictly positive so the low-flux region stays non-degenerate. True near-zero *local* flux needs more than a bound relaxation — see the forward-note.
- **Homogeneous across all six** `Prior_specs/*_priors.R` — the intercomparison requires identical construction.
- **Re-calibration required.** Full six-model re-run (Roihu); afterward, check  $\hat{R}$ /ESS for `sigma_input` specifically on TP2/TP3 to confirm the clean mixing the re-parameterisation is meant to buy.

## 7 Forward-note: toward a local / stratified sigma

A later step may let `sigma` vary by stratum (or plot) rather than as one national scalar — so that, e.g., recently clear-cut strata can take a genuinely low input. The flux floor is set low (0.05) now to leave that door open, but two things must be handled *before* near-zero local flux is usable, and neither is solved by the bound relaxation alone.

1. **The re-parameterisation removes a *prior* wall, not a *likelihood* cliff.** As the effective flux  $\rightarrow 0$ , the steady-state initial carbon  $C_{\text{init}} = J/(\alpha \xi) \rightarrow 0$ , so predicted SOC  $\rightarrow 0$ . Under the multiplicative-normal error the standard deviation ( $\propto$  predicted SOC) also  $\rightarrow 0$ , and for a plot with real, non-zero observed SOC the log-likelihood diverges to  $-\infty$ . That is a cliff in the *likelihood*; the smooth ruler does nothing about it. For the global run it is harmless (no posterior mass down there); for a low-flux stratum it becomes live.
2. **Steady-state initialisation is wrong for a disturbed plot.** A clear-cut removes *inputs*, not the existing *soil stock* (soil C decays slowly). But steady-state init ties  $C_{\text{init}}$  to the current



(low) flux, forcing near-zero initial carbon and clashing with the observed stock. Local low-flux calibration therefore needs a non-steady-state initial condition for disturbed plots, and likely an **additive error floor** (or a floor on the sd) so the likelihood cannot collapse.

These are modelling changes, logged here for when the stratified step is taken.

## 8 Decisions taken

1. Scope = **boreal**, at the **widest defensible** end (Gower 2001 NPP max), not European/global (a tropical ceiling would not usefully constrain even TP2; a boreal ceiling is the physically correct one for a Finnish application).
2. Bound the **effective flux**, not the multiplier, so it is homogeneous in physical units across models with differing  $J$ .
3. **Floor relaxed to 0.05 tC/ha/yr** (below the Gower minimum 0.52) — a strictly-positive, non-binding guardrail. Keeps SP1’s sub-1 fit admissible, stays non-degenerate (not zero), and pre-positions for a future local/stratified **sigma**.
4. Enforcement mechanism = **re-parameterisation** (see §“How to enforce the bound in the sampler”). Rather than a hard wall in the prior — which mixes badly exactly for the binding models TP2/TP3, since their **sigma\_input** sits against the ceiling — the parameter is sampled on an unbounded, stretched coordinate that reads off a value inside the window by construction: same posterior, but no wall for the sampler to reject against. This supersedes the note’s earlier “truncated-lognormal” idea.
5. **Yasso20 stays free** (widest envelope;  $7.5 < 8.7$ ) — only TP2/TP3 bind.
6. **Uniform construction across the two auxiliary parameters**, via a joint, flux-space re-parameterisation: both the contemporary and the 1917 flux are bounded to the *same* Gower envelope, and **sigma\_init** is recovered as their ratio rather than bounded independently. Corrects the earlier (incorrect) claim that independent bounds keep the pre-run flux in-window.
7. **Local/stratified sigma deferred** but pre-positioned (low floor); its prerequisites (non-steady-state init for disturbed plots, additive error floor) are logged in the forward-note.

## 9 Appendix: the construction in full (deeper reading layer)

This appendix formalises what the narrative above described in words: the bounded transform, the two-flux re-parameterisation, and how both slot into the sampler. Nothing here changes the plain-language account — it is the same object written algebraically, for implementation and review. A reader happy with the “endless ruler” picture can stop at the previous section.

### 9.1 A.1 The bounded (scaled-logit) transform

Let  $z \in \mathbb{R}$  be an unconstrained coordinate the sampler moves freely, and  $[a, b]$  the physical window. The map onto the window is a scaled logistic:

$$p = g(z) = a + (b - a) \sigma(z), \quad \sigma(z) = \frac{1}{1 + e^{-z}} \in (0, 1),$$

so  $g : \mathbb{R} \rightarrow (a, b)$ , with  $g(-\infty) = a$ ,  $g(0) = \frac{a+b}{2}$  and  $g(+\infty) = b$ . This is the “endless ruler”: the whole real line is mapped into the finite window, the two edges sitting at  $z = \mp\infty$ . Its inverse (physical  $\rightarrow$  sampler coordinate) is

$$z = g^{-1}(p) = \log \frac{p - a}{b - p},$$

the logit of the *fractional position* of  $p$  within the window — literally the engine’s existing `logit` transform composed with an affine rescale.

The local stretch factor (the one-dimensional Jacobian) is the derivative

$$\frac{dp}{dz} = (b-a)\sigma(z)(1-\sigma(z)) = \frac{(p-a)(b-p)}{b-a},$$

maximal at the centre  $((b-a)/4)$  and vanishing *linearly* toward each end ( $\propto p-a$  near the floor,  $\propto b-p$  near the ceiling) — which is exactly why the edges are approached but never reached. The log-Jacobian, the term added to the log-prior, is

$$\log \left| \frac{dp}{dz} \right| = \log(p-a) + \log(b-p) - \log(b-a).$$

For any chosen prior density  $\pi_p(p)$  on the physical value, the induced density in the sampling coordinate is  $\pi_z(z) = \pi_p(g(z)) |g'(z)|$ ; in log-form the sampler accumulates  $\log \pi_p(p) + \log |g'(z)|$ . A **flat** prior on  $p$  across  $[a, b]$  therefore contributes just the log-Jacobian (up to a constant); a mild centre-pull, if wanted, enters through  $\log \pi_p$ . The Jacobian is what makes “flat on  $p$ ” actually flat, rather than an accidental by-product of the stretch.

## 9.2 A.2 The two-flux re-parameterisation

Two physical fluxes must stay inside the envelope  $[a, b] = [0.05, 8.7]$ : the contemporary effective input  $F_{\text{now}}$  and the 1917 pre-run input  $F_{1917}$ . Give each its own unconstrained coordinate and map both through the *same* bounded transform:

$$F_{\text{now}} = g(z_1; a, b), \quad F_{1917} = g(z_2; a, b).$$

The physical multipliers the wrappers expect are then recovered as

$$\text{sigma\_input} = \frac{F_{\text{now}}}{\bar{J}}, \quad \text{sigma\_init} = \frac{F_{1917}}{F_{\text{now}}},$$

with  $\bar{J}$  the frozen mean-litter units-bridge (§“A scale factor...”). This reproduces the model’s existing algebra: the wrapper forms  $J_{1985} = \text{sigma\_input} \times J_{t_0} \approx F_{\text{now}}$  and  $J_{1917} = \text{sigma\_init} \times \text{sigma\_input} \times J_{\text{full}} \approx F_{1917}$ , so **both fluxes are bounded by construction** and `sigma_init` inherits the range  $[a/b, b/a] = [0.006, 174]$  — wide, as it should be (we have no direct information on 1917 productivity beyond the envelope), with the SOC data identifying it within (posteriors 0.19–0.72).

Because the transform is separable in  $(z_1, z_2)$ , the joint log-prior is just the sum of the two independent pieces,

$$\log \pi(z_1, z_2) = \underbrace{\log \pi_p(F_{\text{now}}) + \log |g'(z_1)|}_{\text{contemporary}} + \underbrace{\log \pi_p(F_{1917}) + \log |g'(z_2)|}_{\text{historical}},$$

each term exactly as in A.1; the Jacobian is diagonal, no cross terms.

### 9.3 A.3 Where it lives in the engine

The transform machinery is `build_transforms()` in the base engine. Each parameter group carries a `type`; the existing diagonal types (`log`, `logit`, `unconstrained`) dispatch inside `to_original`, `to_unconstrained` and `log_jacobian`. The bounded map is **one new type**, `bounded`, carrying its window  $(a, b)$ :

```
to_original      : p  <- a + (b - a) / (1 + exp(-z))
to_unconstrained : z  <- log((p - a) / (b - p))
log_jacobian     : jac <- jac + sum( log(p - a) + log(b - p) - log(b - a) )
```

$F_{\text{now}}$  and  $F_{1917}$  are two `bounded` coordinates; the recovery of `sigma_input/sigma_init` (the divisions in A.2) happens once per likelihood call, either at the tail of `to_original` or the head of the wrapper, using the frozen  $\bar{J}$ . The window  $(a, b)$  is stated in **physical flux units** in each `Prior_specs/*_priors.R`;  $\bar{J}$  is computed once at setup from the litter the engine already holds. No per-iteration data read is added — the run-time path is identical to the current one, plus two cheap evaluations of  $g$  and their log-Jacobians.

### 9.4 A.4 The per-iteration path, end to end

```
sampler proposes z = (... , z1, z2, ...)      # unconstrained, wall-free
  F_now = g(z1; a, b)                        # bounded transform
  F_1917 = g(z2; a, b)
  sigma_input = F_now / Jbar                  # recover multipliers
  sigma_init = F_1917 / F_now
  -> model_params
wrapper: J_1985 = sigma_input * J_t0          # data (litter) enters here, as now
        J_1917 = sigma_init * sigma_input * J_full
run model -> SOC_hat -> likelihood
log_posterior += log_prior(F_now) + logJ(z1)
                + log_prior(F_1917) + logJ(z2)    # Jacobian keeps priors honest
```

Every proposal  $z$  is feasible, both fluxes lie inside  $[a, b]$  by construction, and the two log-Jacobian terms guarantee the physical-space priors are the ones we intended. That is the whole mechanism.

## 10 References

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